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Recoverable AI Risk Evidence

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Table of Contents

Executive Summary	1
Part 1: Foundational Information	4
1 Framing Risk	4
1.1 Understanding and Addressing Risks, Impacts, and Harms	4
1.2 Challenges for AI Risk Management	5
1.2.1 Risk Measurement	5
1.2.2 Risk Tolerance	7
1.2.3 Risk Prioritization	7
1.2.4 Organizational Integration and Management of Risk	8
2 Audience	9
3 AI Risks and Trustworthiness	12
3.1 Valid and Reliable	13
3.2 Safe	14
3.3 Secure and Resilient	15
3.4 Accountable and Transparent	15
3.5 Explainable and Interpretable	16
3.6 Privacy-Enhanced	17
3.7 Fair – with Harmful Bias Managed	17
4 Effectiveness of the AI RMF	19
Part 2: Core and Profiles	20
5 AI RMF Core	20
5.1 Govern	21
5.2 Map	24
5.3 Measure	28
5.4 Manage	31
6 AI RMF Profiles	33
Appendix A: Descriptions of AI Actor Tasks from Figures 2 and 3	35
Appendix B: How AI Risks Differ from Traditional Software Risks	38
Appendix C: AI Risk Management and Human-AI Interaction	40
Appendix D: Attributes of the AI RMF	42

List of Tables

Table 1 Categories and subcategories for the GOVERN function.	22
Table 2 Categories and subcategories for the MAP function.	26
Table 3 Categories and subcategories for the MEASURE function.	29
Table 4 Categories and subcategories for the MANAGE function.	32

3. AI Risks and Trustworthiness

For AI systems to be trustworthy, they often need to be responsive to a multiplicity of criteria that are of value to interested parties. Approaches which enhance AI trustworthiness can reduce negative AI risks. This Framework articulates the following **characteristics** of trustworthy AI and offers guidance for addressing them. Characteristics of trustworthy AI systems include: **valid and reliable, safe, secure and resilient, accountable and transparent, explainable and interpretable, privacy-enhanced, and fair with harmful bias managed**. Creating trustworthy AI requires balancing each of these characteristics based on the AI system's context of use. While all characteristics are socio-technical system attributes, accountability and transparency also relate to the processes and activities internal to an AI system and its external setting. Neglecting these characteristics can increase the probability and magnitude of negative consequences.



Fig. 4. Characteristics of trustworthy AI systems. Valid & Reliable is a necessary condition of trustworthiness and is shown as the base for other trustworthiness characteristics. Accountable & Transparent is shown as a vertical box because it relates to all other characteristics.

Trustworthiness characteristics (shown in Figure 4) are inextricably tied to social and organizational behavior, the datasets used by AI systems, selection of AI models and algorithms and the decisions made by those who build them, and the interactions with the humans who provide insight from and oversight of such systems. Human judgment should be employed when deciding on the specific metrics related to AI trustworthiness characteristics and the precise threshold values for those metrics.

Addressing AI trustworthiness characteristics individually will not ensure AI system trustworthiness; tradeoffs are usually involved, rarely do all characteristics apply in every setting, and some will be more or less important in any given situation. Ultimately, trustworthiness is a social concept that ranges across a spectrum and is only as strong as its weakest characteristics.

When managing AI risks, organizations can face difficult decisions in balancing these characteristics. For example, in certain scenarios tradeoffs may emerge between optimizing for interpretability and achieving privacy. In other cases, organizations might face a tradeoff between predictive accuracy and interpretability. Or, under certain conditions such as data sparsity, privacy-enhancing techniques can result in a loss in accuracy, affecting decisions

GOVERN is a cross-cutting function that is infused throughout AI risk management and enables the other functions of the process. Aspects of **GOVERN**, especially those related to compliance or evaluation, should be integrated into each of the other functions. Attention to governance is a continual and intrinsic requirement for effective AI risk management over an AI system's lifespan and the organization's hierarchy.

Strong governance can drive and enhance internal practices and norms to facilitate organizational risk culture. Governing authorities can determine the overarching policies that direct an organization's mission, goals, values, culture, and risk tolerance. Senior leadership sets the tone for risk management within an organization, and with it, organizational culture. Management aligns the technical aspects of AI risk management to policies and operations. Documentation can enhance transparency, improve human review processes, and bolster accountability in AI system teams.

After putting in place the structures, systems, processes, and teams described in the **GOVERN** function, organizations should benefit from a purpose-driven culture focused on risk understanding and management. It is incumbent on Framework users to continue to execute the **GOVERN** function as knowledge, cultures, and needs or expectations from AI actors evolve over time.

Practices related to governing AI risks are described in the NIST AI RMF Playbook. Table 1 lists the **GOVERN** function's categories and subcategories.

Table 1: Categories and subcategories for the **GOVERN** function.

Categories	Subcategories
GOVERN 1: Policies, processes, procedures, and practices across the organization related to the mapping, measuring, and managing of AI risks are in place, transparent, and implemented effectively.	GOVERN 1.1: Legal and regulatory requirements involving AI are understood, managed, and documented. GOVERN 1.2: The characteristics of trustworthy AI are integrated into organizational policies, processes, procedures, and practices. GOVERN 1.3: Processes, procedures, and practices are in place to determine the needed level of risk management activities based on the organization's risk tolerance. GOVERN 1.4: The risk management process and its outcomes are established through transparent policies, procedures, and other controls based on organizational risk priorities.

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Table 1: Categories and subcategories for the **GOVERN** function. (Continued)

Categories	Subcategories
	GOVERN 1.5: Ongoing monitoring and periodic review of the risk management process and its outcomes are planned and organizational roles and responsibilities clearly defined, including determining the frequency of periodic review. GOVERN 1.6: Mechanisms are in place to inventory AI systems and are resourced according to organizational risk priorities. GOVERN 1.7: Processes and procedures are in place for decommissioning and phasing out AI systems safely and in a manner that does not increase risks or decrease the organization's trustworthiness.
GOVERN 2: Accountability structures are in place so that the appropriate teams and individuals are empowered, responsible, and trained for mapping, measuring, and managing AI risks.	GOVERN 2.1: Roles and responsibilities and lines of communication related to mapping, measuring, and managing AI risks are documented and are clear to individuals and teams throughout the organization. GOVERN 2.2: The organization's personnel and partners receive AI risk management training to enable them to perform their duties and responsibilities consistent with related policies, procedures, and agreements. GOVERN 2.3: Executive leadership of the organization takes responsibility for decisions about risks associated with AI system development and deployment.
GOVERN 3: Workforce diversity, equity, inclusion, and accessibility processes are prioritized in the mapping, measuring, and managing of AI risks throughout the lifecycle.	GOVERN 3.1: Decision-making related to mapping, measuring, and managing AI risks throughout the lifecycle is informed by a diverse team (e.g., diversity of demographics, disciplines, experience, expertise, and backgrounds). GOVERN 3.2: Policies and procedures are in place to define and differentiate roles and responsibilities for human-AI configurations and oversight of AI systems.
GOVERN 4: Organizational teams are committed to a culture	GOVERN 4.1: Organizational policies and practices are in place to foster a critical thinking and safety-first mindset in the design, development, deployment, and uses of AI systems to minimize potential negative impacts.

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Table 1: Categories and subcategories for the **GOVERN** function. (Continued)

Categories	Subcategories
that considers and communicates AI risk.	GOVERN 4.2: Organizational teams document the risks and potential impacts of the AI technology they design, develop, deploy, evaluate, and use, and they communicate about the impacts more broadly. GOVERN 4.3: Organizational practices are in place to enable AI testing, identification of incidents, and information sharing.
GOVERN 5: Processes are in place for robust engagement with relevant AI actors.	GOVERN 5.1: Organizational policies and practices are in place to collect, consider, prioritize, and integrate feedback from those external to the team that developed or deployed the AI system regarding the potential individual and societal impacts related to AI risks. GOVERN 5.2: Mechanisms are established to enable the team that developed or deployed AI systems to regularly incorporate adjudicated feedback from relevant AI actors into system design and implementation.
GOVERN 6: Policies and procedures are in place to address AI risks and benefits arising from third-party software and data and other supply chain issues.	GOVERN 6.1: Policies and procedures are in place that address AI risks associated with third-party entities, including risks of infringement of a third-party's intellectual property or other rights. GOVERN 6.2: Contingency processes are in place to handle failures or incidents in third-party data or AI systems deemed to be high-risk.

5.2 Map

The **MAP** function establishes the context to frame risks related to an AI system. The AI lifecycle consists of many interdependent activities involving a diverse set of actors (See Figure 3). In practice, AI actors in charge of one part of the process often do not have full visibility or control over other parts and their associated contexts. The interdependencies between these activities, and among the relevant AI actors, can make it difficult to reliably anticipate impacts of AI systems. For example, early decisions in identifying purposes and objectives of an AI system can alter its behavior and capabilities, and the dynamics of deployment setting (such as end users or impacted individuals) can shape the impacts of AI system decisions. As a result, the best intentions within one dimension of the AI lifecycle can be undermined via interactions with decisions and conditions in other, later activities.

Table 2: Categories and subcategories for the **MAP** function.

Categories	Subcategories
MAP 1: Context is established and understood.	MAP 1.1: Intended purposes, potentially beneficial uses, context-specific laws, norms and expectations, and prospective settings in which the AI system will be deployed are understood and documented. Considerations include: the specific set or types of users along with their expectations; potential positive and negative impacts of system uses to individuals, communities, organizations, society, and the planet; assumptions and related limitations about AI system purposes, uses, and risks across the development or product AI lifecycle; and related TEVV and system metrics. MAP 1.2: Interdisciplinary AI actors, competencies, skills, and capacities for establishing context reflect demographic diversity and broad domain and user experience expertise, and their participation is documented. Opportunities for interdisciplinary collaboration are prioritized. MAP 1.3: The organization’s mission and relevant goals for AI technology are understood and documented. MAP 1.4: The business value or context of business use has been clearly defined or – in the case of assessing existing AI systems – re-evaluated. MAP 1.5: Organizational risk tolerances are determined and documented. MAP 1.6: System requirements (e.g., “the system shall respect the privacy of its users”) are elicited from and understood by relevant AI actors. Design decisions take socio-technical implications into account to address AI risks.
MAP 2: Categorization of the AI system is performed.	MAP 2.1: The specific tasks and methods used to implement the tasks that the AI system will support are defined (e.g., classifiers, generative models, recommenders). MAP 2.2: Information about the AI system’s knowledge limits and how system output may be utilized and overseen by humans is documented. Documentation provides sufficient information to assist relevant AI actors when making decisions and taking subsequent actions.

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Table cell	Fine	Row / column	Partial	Metric drift	Header + cell
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Source hash	Version	Document	No	Wrong edition	SHA-256 record

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Practices related to measuring AI risks are described in the NIST AI RMF Playbook. Table 3 lists the **MEASURE** function's categories and subcategories.

Table 3: Categories and subcategories for the **MEASURE** function.

Categories	Subcategories
MEASURE 1: Appropriate methods and metrics are identified and applied.	MEASURE 1.1: Approaches and metrics for measurement of AI risks enumerated during the MAP function are selected for implementation starting with the most significant AI risks. The risks or trustworthiness characteristics that will not – or cannot – be measured are properly documented. MEASURE 1.2: Appropriateness of AI metrics and effectiveness of existing controls are regularly assessed and updated, including reports of errors and potential impacts on affected communities. MEASURE 1.3: Internal experts who did not serve as front-line developers for the system and/or independent assessors are involved in regular assessments and updates. Domain experts, users, AI actors external to the team that developed or deployed the AI system, and affected communities are consulted in support of assessments as necessary per organizational risk tolerance.
MEASURE 2: AI systems are evaluated for trustworthy characteristics.	MEASURE 2.1: Test sets, metrics, and details about the tools used during TEVV are documented. MEASURE 2.2: Evaluations involving human subjects meet applicable requirements (including human subject protection) and are representative of the relevant population. MEASURE 2.3: AI system performance or assurance criteria are measured qualitatively or quantitatively and demonstrated for conditions similar to deployment setting(s). Measures are documented. MEASURE 2.4: The functionality and behavior of the AI system and its components – as identified in the MAP function – are monitored when in production. MEASURE 2.5: The AI system to be deployed is demonstrated to be valid and reliable. Limitations of the generalizability beyond the conditions under which the technology was developed are documented.

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Practices related to managing AI risks are described in the NIST AI RMF Playbook. Table 4 lists the **MANAGE** function's categories and subcategories.

Table 4: Categories and subcategories for the **MANAGE** function.

Categories	Subcategories
MANAGE 1: AI risks based on assessments and other analytical output from the MAP and MEASURE functions are prioritized, responded to, and managed.	MANAGE 1.1: A determination is made as to whether the AI system achieves its intended purposes and stated objectives and whether its development or deployment should proceed. MANAGE 1.2: Treatment of documented AI risks is prioritized based on impact, likelihood, and available resources or methods. MANAGE 1.3: Responses to the AI risks deemed high priority, as identified by the MAP function, are developed, planned, and documented. Risk response options can include mitigating, transferring, avoiding, or accepting. MANAGE 1.4: Negative residual risks (defined as the sum of all unmitigated risks) to both downstream acquirers of AI systems and end users are documented.
MANAGE 2: Strategies to maximize AI benefits and minimize negative impacts are planned, prepared, implemented, documented, and informed by input from relevant AI actors.	MANAGE 2.1: Resources required to manage AI risks are taken into account – along with viable non-AI alternative systems, approaches, or methods – to reduce the magnitude or likelihood of potential impacts. MANAGE 2.2: Mechanisms are in place and applied to sustain the value of deployed AI systems. MANAGE 2.3: Procedures are followed to respond to and recover from a previously unknown risk when it is identified. MANAGE 2.4: Mechanisms are in place and applied, and responsibilities are assigned and understood, to supersede, disengage, or deactivate AI systems that demonstrate performance or outcomes inconsistent with intended use.
MANAGE 3: AI risks and benefits from third-party entities are managed.	MANAGE 3.1: AI risks and benefits from third-party resources are regularly monitored, and risk controls are applied and documented. MANAGE 3.2: Pre-trained models which are used for development are monitored as part of AI system regular monitoring and maintenance.

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Appendix C:

AI Risk Management and Human-AI Interaction

Organizations that design, develop, or deploy AI systems for use in operational settings may enhance their AI risk management by understanding current limitations of human-AI interaction. The AI RMF provides opportunities to clearly define and differentiate the various human roles and responsibilities when using, interacting with, or managing AI systems.

Many of the data-driven approaches that AI systems rely on attempt to convert or represent individual and social observational and decision-making practices into measurable quantities. Representing complex human phenomena with mathematical models can come at the cost of removing necessary context. This loss of context may in turn make it difficult to understand individual and societal impacts that are key to AI risk management efforts.

Issues that merit further consideration and research include:

1. **Human roles and responsibilities in decision making and overseeing AI systems need to be clearly defined and differentiated.** Human-AI configurations can span from fully autonomous to fully manual. AI systems can autonomously make decisions, defer decision making to a human expert, or be used by a human decision maker as an additional opinion. Some AI systems may not require human oversight, such as models used to improve video compression. Other systems may specifically require human oversight.
2. **Decisions that go into the design, development, deployment, evaluation, and use of AI systems reflect systemic and human cognitive biases.** AI actors bring their cognitive biases, both individual and group, into the process. Biases can stem from end-user decision-making tasks and be introduced across the AI lifecycle via human assumptions, expectations, and decisions during design and modeling tasks. These biases, which are not necessarily always harmful, may be exacerbated by AI system opacity and the resulting lack of transparency. Systemic biases at the organizational level can influence how teams are structured and who controls the decision-making processes throughout the AI lifecycle. These biases can also influence downstream decisions by end users, decision makers, and policy makers and may lead to negative impacts.
3. **Human-AI interaction results vary.** Under certain conditions – for example, in perceptual-based judgment tasks – the AI part of the human-AI interaction can amplify human biases, leading to more biased decisions than the AI or human alone. When these variations are judiciously taken into account in organizing human-AI teams, however, they can result in complementarity and improved overall performance.

Appendix A:

Descriptions of AI Actor Tasks from Figures 2 and 3

AI Design tasks are performed during the Application Context and Data and Input phases of the AI lifecycle in Figure 2. AI Design actors create the concept and objectives of AI systems and are responsible for the planning, design, and data collection and processing tasks of the AI system so that the AI system is lawful and fit-for-purpose. Tasks include articulating and documenting the system's concept and objectives, underlying assumptions, context, and requirements; gathering and cleaning data; and documenting the metadata and characteristics of the dataset. AI actors in this category include data scientists, domain experts, socio-cultural analysts, experts in the field of diversity, equity, inclusion, and accessibility, members of impacted communities, human factors experts (e.g., UX/UI design), governance experts, data engineers, data providers, system funders, product managers, third-party entities, evaluators, and legal and privacy governance.

AI Development tasks are performed during the AI Model phase of the lifecycle in Figure 2. AI Development actors provide the initial infrastructure of AI systems and are responsible for model building and interpretation tasks, which involve the creation, selection, calibration, training, and/or testing of models or algorithms. AI actors in this category include machine learning experts, data scientists, developers, third-party entities, legal and privacy governance experts, and experts in the socio-cultural and contextual factors associated with the deployment setting.

AI Deployment tasks are performed during the Task and Output phase of the lifecycle in Figure 2. AI Deployment actors are responsible for contextual decisions relating to how the AI system is used to assure deployment of the system into production. Related tasks include piloting the system, checking compatibility with legacy systems, ensuring regulatory compliance, managing organizational change, and evaluating user experience. AI actors in this category include system integrators, software developers, end users, operators and practitioners, evaluators, and domain experts with expertise in human factors, socio-cultural analysis, and governance.

Operation and Monitoring tasks are performed in the Application Context/Operate and Monitor phase of the lifecycle in Figure 2. These tasks are carried out by AI actors who are responsible for operating the AI system and working with others to regularly assess system output and impacts. AI actors in this category include system operators, domain experts, AI designers, users who interpret or incorporate the output of AI systems, product developers, evaluators and auditors, compliance experts, organizational management, and members of the research community.

Test, Evaluation, Verification, and Validation (TEVV) tasks are performed throughout the AI lifecycle. They are carried out by AI actors who examine the AI system or its components, or detect and remediate problems. Ideally, AI actors carrying out verification

Appendix B:

How AI Risks Differ from Traditional Software Risks

As with traditional software, risks from AI-based technology can be bigger than an enterprise, span organizations, and lead to societal impacts. AI systems also bring a set of risks that are not comprehensively addressed by current risk frameworks and approaches. Some AI system features that present risks also can be beneficial. For example, pre-trained models and transfer learning can advance research and increase accuracy and resilience when compared to other models and approaches. Identifying contextual factors in the **MAP** function will assist AI actors in determining the level of risk and potential management efforts.

Compared to traditional software, AI-specific risks that are new or increased include the following:

- The data used for building an AI system may not be a true or appropriate representation of the context or intended use of the AI system, and the ground truth may either not exist or not be available. Additionally, harmful bias and other data quality issues can affect AI system trustworthiness, which could lead to negative impacts.
- AI system dependency and reliance on data for training tasks, combined with increased volume and complexity typically associated with such data.
- Intentional or unintentional changes during training may fundamentally alter AI system performance.
- Datasets used to train AI systems may become detached from their original and intended context or may become stale or outdated relative to deployment context.
- AI system scale and complexity (many systems contain billions or even trillions of decision points) housed within more traditional software applications.
- Use of pre-trained models that can advance research and improve performance can also increase levels of statistical uncertainty and cause issues with bias management, scientific validity, and reproducibility.
- Higher degree of difficulty in predicting failure modes for emergent properties of large-scale pre-trained models.
- Privacy risk due to enhanced data aggregation capability for AI systems.
- AI systems may require more frequent maintenance and triggers for conducting corrective maintenance due to data, model, or concept drift.
- Increased opacity and concerns about reproducibility.
- Underdeveloped software testing standards and inability to document AI-based practices to the standard expected of traditionally engineered software for all but the simplest of cases.
- Difficulty in performing regular AI-based software testing, or determining what to test, since AI systems are not subject to the same controls as traditional code development.

4. **Presenting AI system information to humans is complex.** Humans perceive and derive meaning from AI system output and explanations in different ways, reflecting different individual preferences, traits, and skills.

The **GOVERN** function provides organizations with the opportunity to clarify and define the roles and responsibilities for the humans in the Human-AI team configurations and those who are overseeing the AI system performance. The **GOVERN** function also creates mechanisms for organizations to make their decision-making processes more explicit, to help counter systemic biases.

The **MAP** function suggests opportunities to define and document processes for operator and practitioner proficiency with AI system performance and trustworthiness concepts, and to define relevant technical standards and certifications. Implementing **MAP** function categories and subcategories may help organizations improve their internal competency for analyzing context, identifying procedural and system limitations, exploring and examining impacts of AI-based systems in the real world, and evaluating decision-making processes throughout the AI lifecycle.

The **GOVERN** and **MAP** functions describe the importance of interdisciplinarity and demographically diverse teams and utilizing feedback from potentially impacted individuals and communities. AI actors called out in the AI RMF who perform human factors tasks and activities can assist technical teams by anchoring in design and development practices to user intentions and representatives of the broader AI community, and societal values. These actors further help to incorporate context-specific norms and values in system design and evaluate end user experiences – in conjunction with AI systems.

AI risk management approaches for human-AI configurations will be augmented by ongoing research and evaluation. For example, the degree to which humans are empowered and incentivized to challenge AI system output requires further studies. Data about the frequency and rationale with which humans overrule AI system output in deployed systems may be useful to collect and analyze.

Appendix C: provenance and page map

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